

Tony Zhu

(587)-589-9671 | t96zhu@uwaterloo.ca | www.linkedin.com/in/tony-zhu-393122367 | <https://websiteah.vercel.app/>

EDUCATION

University of Waterloo

Bachelor of Applied Science in Electrical Engineering

Waterloo, ON

Present

Harry Ainlay Highschool

International Baccalareate Certificate

Edmonton, AB

Sept. 2021 – June 2025

TECHNICAL SKILLS

Languages: C, C++, Python

Hardware & EDA: Altium Designer, PCB Layout, Schematic Capture, Arduino IDE, AutoCAD

Developer Tools: VS Code, CLion, Git, Powershell

Productivity: Microsoft Office, Google Workspace

PROJECTS

Real-Time Embedded PC Monitor | C++, Python, Serial Comms

Jan. 2026

- Engineered a hardware-based telemetry system to visualize real-time PC metrics (CPU, RAM, Ping) on an LCD1602 display.
- Developed a Python background service using `psutil` and `subprocess` to harvest system data with < 0.1% CPU overhead.
- Wrote C++ firmware implementing circular buffer parsing to decode serial data packets and dynamically update the interface.

Ultrasonic Distance Security System | C++, Arduino, Circuit Design

Dec. 2025

- Designed and prototyped a security system integrating an HC-SR04 ultrasonic sensor, LCD 1602 display, and IR receiver to detect objects within a 2-400cm range.
- Implemented a C++ finite state machine to manage distinct "Active" and "Silent" operating modes, toggleable via a decoded IR remote signal.
- Optimized system performance by utilizing interrupt handling to resolve timing conflicts between the IR receiver and LCD refresh cycles.
- Engineered a multi-stage alert system with visual LED indicators and frequency-modulated audio alarms based on real-time proximity data.

Wifi/Website Controlled Lighting System | Arduino, C++

Oct. 2025 – Nov. 2025

- Designed and programmed a two-node IoT system using ESP8266 microcontrollers to enable wireless communication over Wi-Fi.
- Configured one ESP as a transmitter to send RGB LED control signals to a receiver node hosting a web server.
- Implemented HTTP-based communication for real-time color and flashing control.
- Developed and debugged embedded firmware using the Arduino framework and serial monitoring tools.

EXPERIENCE

Design Team Member

Jan. 2026 – Present

Waterloo Aerial Robotics Group (WARG)

Waterloo, ON

- Selected for the Electrical sub-team after successfully completing a rigorous PCB design bootcamp using Altium Designer.
- Executed the full hardware design flow including schematic capture, component selection, and PCB layout in compliance with team standards.
- Collaborating with the hardware team to assemble, solder, and validate custom electronics for autonomous aerial vehicles.

Badminton Coach

July 2023 – Present

The Derrick Golf and Winter Club | JA Badminton

Edmonton, AB

- Assisted in training athletes ages 6-16, focusing on skill development, discipline, and teamwork.
- Provided one-on-one feedback and mentorship, improving athlete performance in competitions.
- Organized practices, drills, and match strategies, developing leadership skills, and event coordination.